

Практикум C6:archive

Twenty-two questions, 123 marks. Every question the archive asks about planes, from May 2021 to November 2025, in the order of the seven techniques rather than the order of the papers.

О практикуме

Разрез продолжает C5: всё, где плоскость — сам предмет вопроса. Уравнение и нормаль, встреча прямой с плоскостью, прямая пересечения двух плоскостей, три плоскости и системы уравнений, перпендикуляр из точки и отражение. 38 баллов из 182 geometry.vectors (0.209) и 98 из 216 geometry.vectors_3d (0.4537), разобрано по-блочному. 136 баллов по корпусу, 123 рабочих — ноябрь 2023 Q12(b)–(d) стоит дважды. 51 балл Paper 1, 72 Paper 2.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-c6-planes

Собрано из ноутбука репозитория [ib-math-aa-hl-past-papers](#). Проверки в PDF не работают — они для интерактивной версии.

C6 archive — planes, all of it

Twenty-two questions, 123 marks.

Every question the archive asks about planes, from May 2021 to November 2025, in the order of the seven techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-c6-planes.ipynb`.

§	technique	questions	marks
1	A point and a normal	2	12
2	A normal from two directions	8	29
3	A line and a plane	3	11
4	Two planes	3	10
5	Three planes and systems	4	16
6	The perpendicular: foot and distance	4	29
7	The reflection	3	16

The checks are the same ones the practicum uses and they store nothing:

each is handed the planes and lines exactly as the question prints them — `plane(Eq(2*x - y + z, 4))`, `plane(A, B, C)`, `line(vec(0, -2, 0) + lam*vec(1, 1, -1))` — and a point or a letter the question asks for is handed **the conditions** that fix it: `on(C, P)`, `perpendicular(C - B, P)`, `reflection(image, B, P)`, `meet_in_line(P1, P2, P3)`. A plane is accepted as any multiple of its equation or in vector form; a line with any point on it and any parallel direction. Three significant figures, unless the question asks otherwise.

Parts of one question are split by technique. November 2021 Q11 appears in §§ 2, 3, 4 and 7; each part repeats what it needs.

A *show that* or *verify* is checked by the line before the given result: the left-hand side at the point, $\mathbf{r} \cdot \mathbf{n}$ at the general point of a line, the normal before it is compared with the printed equation.

Enter a plane as `Eq(a*x + b*y + c*z, d)`
and a line as `vec(point) + lam*vec(direction)`; `lam` and `mu` are λ and μ .

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('.')          # from
practicum/geometry to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
vec, plane, cross, intersection, unknown

language('en')                # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

§ 1. A point and a normal

Two questions, 12 marks. A point on a plane, parallel and perpendicular planes, where a plane meets the axes, a line along the normal.

1.1 — May 2024 TZ2 Paper 1 Q11(a)-(c), 8 marks

The plane P_1 has equation $2x + 6y - 2z = 5$.

(a) Verify that the point $A(2, \frac{1}{2}, 1)$ lies on the plane P_1 .

The plane P_2 is given by $(k^2 - 6)x + (2k + 3)y + pz = q$, where $p, q, k \in \mathbb{R}$ and $p \neq 0$.

(b) In the case where $p = -6$, P_2 is perpendicular to P_1 and A lies on P_2 . Find the value of k and the value of q .

For parts (c), (d) and (e) it is now given that P_2 is parallel to P_1 with $k = 3$.

(c) Determine the value of p .

For (a) enter the left-hand side of P_1 worked out at A .

```

k, p, q = symbols('k p q')
A = vec(2, Rational(1, 2), 1)
P1 = plane(Eq(2*x + 6*y - 2*z, 5), name='P1')
P2 = plane(Eq((k**2 - 6)*x + (2*k + 3)*y +
p*z, q), name='P2')

q1_la = ...      # 2x + 6y - 2z at A
q1_lb = [...]    # [k, q]
q1_lc = ...      # p

verify_find('1.1(a)', q1_la, P1.side(A))
verify_find('1.1(b)', q1_lb, [k, q],
[perpendicular(P1, P2.subs(p, -6)), on(A,
P2.subs(p, -6))])
verify_find('1.1(c)', q1_lc, p, [parallel(P1,
P2.subs(k, 3))])

```

1.2 — November 2025 TZ1 Paper 2 Q12(c), (e), 4 marks

The plane Π has Cartesian equation $4x + y - 3z = 18$. The lines L_1 and L_2 lie in Π and intersect at the point with position

vector $\begin{pmatrix} 1 \\ 8 \\ -2 \end{pmatrix}$.

The plane intersects the coordinate axes at $P(4.5, 0, 0)$, $Q(0, q, 0)$ and $R(0, 0, r)$.

(c) Write down the value of **(i)** q ; **(ii)** r .

Another line, L_3 , is normal to Π and passes through the point of intersection of L_1 and L_2 .

(e) Write down an equation for L_3 in the form $\mathbf{r}_3 = \mathbf{m} + \gamma \mathbf{n}$.

Use `lam` for γ .

```
Pi = plane(Eq(4*x + y - 3*z, 18))
X = vec(1, 8, -2)
at_q, at_r = unknown('q'), unknown('r')

q1_2c_i = ...      # q
q1_2c_ii = ...     # r
q1_2e = ...        # L3: vec(...) +
lam*vec(...)

verify_find('1.2(c)(i)', q1_2c_i, at_q,
[on(vec(0, at_q, 0), Pi)])
verify_find('1.2(c)(ii)', q1_2c_ii, at_r,
[on(vec(0, 0, at_r), Pi)])
verify_line('1.2(e)', q1_2e, line(X,
Pi.normal))
```

§ 2. A normal from two directions

Eight questions, 29 marks. Three points, a line and a point, two lines, a plane perpendicular to two planes — and the vector product that turns two directions into a normal.

2.1 — *May 2021 TZ1 Paper 2 Q6(a), 3 marks*

Consider the planes Π_1 and Π_2 with the following equations.

$$\begin{aligned}\Pi_1 : 3x + 2y + z &= 6 & \Pi_2 : x - 2y + z \\ & & = 4\end{aligned}$$

(a) Find a Cartesian equation of the plane Π_3 which is perpendicular to Π_1 and Π_2 and passes through the origin $(0, 0, 0)$.

```
P1 = plane(Eq(3*x + 2*y + z, 6), name='Π1')
P2 = plane(Eq(x - 2*y + z, 4), name='Π2')

q2_1 = ...          # Π3: Eq(...)

verify_plane('2.1', q2_1, plane(vec(0, 0, 0),
P1, P2))
```

2.2 — November 2021 Paper 2

Q11(a), 7 marks

Three points $A(3, 0, 0)$, $B(0, -2, 0)$ and $C(1, 1, -7)$ lie on the plane Π_1 .

(a) (i) Find the vector $\overrightarrow{AB}$ and the vector $\overrightarrow{AC}$.

(ii) Hence find the equation of Π_1 , expressing your answer in the form $ax + by + cz = d$, where $a, b, c, d \in \mathbb{Z}$.

```
A, B, C = vec(3, 0, 0), vec(0, -2, 0), vec(1, 1, -7)
```

```
q2_2_AB = ...      # AB
```

```
q2_2_AC = ...      # AC
```

```
q2_2 = ...          #  $\Pi_1$ : Eq(...)
```

```
verify_find('2.2(a)(i) AB', q2_2_AB, B - A)
```

```
verify_find('2.2(a)(i) AC', q2_2_AC, C - A)
```

```
verify_plane('2.2(a)(ii)', q2_2, plane(A, B, C))
```

2.3 — May 2023 TZ1 Paper 1 Q12(d), 2 marks

Two lines, L_1 and L_2 , intersect at point P . Point $A(2t, 8, 3)$, where $t > 0$, lies on L_2 .

The direction vector of L_1 is $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$, and

$\overrightarrow{PA} = \begin{pmatrix} 2t \\ 0 \\ 3+t \end{pmatrix}$. It is found in part (b) that $t = 3$.

A plane, Π , contains L_1 and L_2 .

(d) Find a normal vector to Π .

Parts (a)–(c) and (e) are in C5, A4, C7 and C2. P is not given, and a normal does not need it: the check puts P at the origin.

```
PA = vec(6, 0, 6) #
(2t, 0, 3 + t) at t = 3
point_P = vec(0, 0, 0) #
not given; a normal does not depend on it
Pi = plane(line(point_P, vec(1, 1, 0)),
line(point_P, PA))

q2_3 = ... # a normal to Π

verify_normal_vector('2.3', q2_3, Pi)
```

2.4 — May 2025 TZ1 Paper 1 Q11(b), 4 marks

The plane P_1 has equation $x + 2y + z = 0$ and the plane P_2 has equation $x - y - 2z = 0$.

A third plane P_3 is perpendicular to both P_1 and P_2 .

The unique point of intersection of all three planes is the point $R(5, -5, 5)$.

(b) Find the Cartesian equation of P_3 .

```
P1 = plane(Eq(x + 2*y + z, 0), name='P1')
P2 = plane(Eq(x - y - 2*z, 0), name='P2')

q2_4 = ...          # P3: Eq(...)

verify_plane('2.4', q2_4, plane(vec(5, -5,
5), P1, P2))
```

2.5 — May 2025 TZ2 Paper 2 Q9(a), 4 marks

A line L_1 has vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ where $t \in \mathbb{R}$.

The plane P_1 contains the line L_1 and passes through the point $(2, 1, 5)$.

(a) Show that the Cartesian equation of the plane P_1 is $x + y - z = -2$.

Enter the normal you get from the line and the point.

```
L1 = line(vec(0, 0, 2) + t*vec(1, 0, 1), t)

q2_5 = ...          # a normal of P1

verify_normal_vector('2.5', q2_5, plane(L1,
vec(2, 1, 5)))
```

2.6 — May 2025 TZ3 Paper 1 Q11(d), 2 marks

The points $A(1, -4, 0)$, $B(-3, -6, 2)$, $C(-1, -2, 4)$ and $D(3, 0, 2)$ form a parallelogram, $ABCD$. It is given that

$$\overrightarrow{AB} \times \overrightarrow{AD} = m \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}, \text{ where } m \in \mathbb{Z}^+.$$

The plane, P_1 , contains the parallelogram $ABCD$.

(d) Find the Cartesian equation of P_1 .

```
A, B, C, D = vec(1, -4, 0), vec(-3, -6, 2),
vec(-1, -2, 4), vec(3, 0, 2)

q2_6 = ...          # P1: Eq(...)

verify_plane('2.6', q2_6, plane(A, B, D))
```

2.7 — November 2025 TZ1 Paper 2
Q12(b), 3 marks

The equations of two lines, L_1 and L_2 , are given by:

$$\begin{aligned} L_1 : \mathbf{r}_1 &= \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \quad \text{where } s \in \mathbb{R} & L_2 : \frac{x+1}{2} \\ &= y - 7 \\ &= \frac{z+5}{3} \end{aligned}$$

The plane Π contains the lines L_1 and L_2 .

(b) (i) Write down the equation of Π , giving your answer in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} + \mu\mathbf{c}$ where $\lambda, \mu \in \mathbb{R}$.

(ii) Given that $\mathbf{b} \times \mathbf{c} = \begin{pmatrix} -4 \\ -1 \\ 3 \end{pmatrix}$, show that the Cartesian equation of Π is $4x + y - 3z = 18$.

For (b)(ii) enter $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.

```

s = symbols('s')
L1 = line(vec(5, 4, 2) + s*vec(1, -1, 1), s)
L2 = cartesian((x + 1)/2, y - 7, (z + 5)/3)
a = vec(5, 4, 2)

q2_7_i = ...      # vec(a) + lam*vec(b) +
mu*vec(c)
q2_7_ii = ...      # a · (b × c)

verify_plane('2.7(b)(i)', q2_7_i, plane(L1,
L2))
verify_find('2.7(b)(ii)', q2_7_ii, dot(a,
vec(-4, -1, 3)))

```

2.8 — November 2025 TZ3 Paper 1 Q10(f), 4 marks

The point $P(-1, 1, -13)$ lies on the line L_1 . The point N on the line L_2 is the point of L_2 closest to P , and it is found in part (e) that $N\left(\frac{1}{3}, -\frac{10}{3}, \frac{7}{3}\right)$.

(f) Point O denotes the origin $(0, 0, 0)$. Find the equation of the plane containing points O , P and N , giving your answer in the form $\alpha x + \beta y + \gamma z = \delta$, where $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$.

```
0, point_P, N = vec(0, 0, 0), vec(-1, 1,
-13), vec(Rational(1, 3), Rational(-10, 3),
Rational(7, 3))
```

```
q2_8 = ...          # the plane OPN: Eq(...)
```

```
verify_plane('2.8', q2_8, plane(0, point_P,
N))
```

§ 3. A line and a plane

Three questions, 11 marks. The general point of a line into the equation of a plane: one value of the parameter, every value, or none.

3.1 — November 2021 Paper 2 **Q11(c), 3 marks**

The line L has vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$.

(c) The plane Π_3 is given by $2x - 2z = 3$. The line L and the plane Π_3 intersect at the point P .

(i) Show that at the point P , $\lambda = \frac{3}{4}$.

(ii) Hence find the coordinates of P .

For (i) enter the value of λ that substituting gives.

```

L = line(vec(0, -2, 0) + lam*vec(1, 1, -1))
P3 = plane(Eq(2*x - 2*z, 3), name='Π3')
at_P = unknown('lambda')

q3_1_i = ...      # λ where L meets Π3
q3_1_ii = ...     # the point where L meets Π3

verify_find('3.1(c)(i)', q3_1_i, at_P,
[on(L.at(at_P), P3)])
verify_intersection('3.1(c)(ii)', q3_1_ii, L,
P3)

```

3.2 — November 2023 TZ1 Paper 2 Q12(b), 3 marks

Line L is given by the vector equation

$$\mathbf{r}_1 = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + s \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \text{ where } s \in \mathbb{R}.$$

Line M is given by the vector equation

$$\mathbf{r}_2 = \begin{pmatrix} 9 \\ 9 \\ 11 \end{pmatrix} + t \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \text{ where } t \in \mathbb{R}.$$

(b) Verify that the lines L and M both lie

in the plane P given by $\mathbf{r} \cdot \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 7$.

Enter $\mathbf{r} \cdot (0, 2, -1)$ at the general point of each line, in terms of its parameter.

```

s = symbols('s')
L = line(vec(1, 2, -3) + s*vec(2, 3, 6), s)
M = line(vec(9, 9, 11) + t*vec(4, 1, 2), t)
n = vec(0, 2, -1)

q3_2_L = ...      # r · (0, 2, -1) along L, in
terms of s
q3_2_M = ...      # the same along M, in terms
of t

verify_find('3.2(b) L', q3_2_L, dot(L.at(s),
n))
verify_find('3.2(b) M', q3_2_M, dot(M.at(t),
n))

```

3.3 — May 2025 TZ3 Paper 1 Q11(f), 5 marks

The plane P_1 has equation $-x + y - z = -5$ and contains the parallelogram $ABCD$, whose diagonals intersect at $E(0, -3, 2)$.

A second plane, P_2 , has Cartesian equation $5x + y - 7z = 1$.

The line L passes through E and is perpendicular to P_1 .

The line L intersects the plane P_2 at point F .

(f) Find the coordinates of F .

Enter the line L too.

```
E = vec(0, -3, 2)
P1 = plane(Eq(-x + y - z, -5), name='P1')
P2 = plane(Eq(5*x + y - 7*z, 1), name='P2')
L = line(E, P1.normal)

q3_3_L = ...      # L: vec(...) + lam*vec(...)
q3_3 = ...        # F

verify_line('3.3(f) L', q3_3_L, L)
verify_intersection('3.3(f)', q3_3, L, P2)
```

§ 4. Two planes

Three questions, 10 marks. The line where two planes meet: a common point and the direction $\mathbf{n}_1 \times \mathbf{n}_2$.

4.1 — November 2021 Paper 2 Q11(b), 2 marks

The plane Π_1 has equation $2x - 3y - z = 6$.
Plane Π_2 has equation $3x - y + 2z = 2$.

(b) The line L is the intersection of Π_1 and Π_2 . Verify that the vector equation of

L can be written as $\mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$.

Enter $\mathbf{n}_1 \times \mathbf{n}_2$.

```
P1 = plane(Eq(2*x - 3*y - z, 6), name='Π1')
P2 = plane(Eq(3*x - y + 2*z, 2), name='Π2')

q4_1 = ...          # n1 × n2

verify_direction('4.1(b)', q4_1,
intersection(P1, P2))
```

4.2 — May 2022 TZ1 Paper 1 Q11(b), 5 marks

Consider the planes $\Pi_1 : 2x - y + z = 4$ and $\Pi_2 : x - 2y + 3z = 5$.

(b) (i) Verify that the point $P(1, -2, 0)$ lies on both Π_1 and Π_2 .

(ii) Find a vector equation of L , the line of intersection of Π_1 and Π_2 .

```
P1 = plane(Eq(2*x - y + z, 4), name='Π1')
P2 = plane(Eq(x - 2*y + 3*z, 5), name='Π2')

q4_2 = ...      # L: vec(...) + lam*vec(...)

verify_intersection('4.2(b)(ii)', q4_2, P1,
P2)
```

4.3 — May 2023 TZ2 Paper 2 Q6(a), 3 marks

Consider the two planes $\Pi_1 : 2x - y + 2z = 6$ and $\Pi_2 : 4x + 3y - z = 2$.

Let L be the line of intersection of Π_1 and Π_2 .

(a) Verify that a vector equation of L is

$$\mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}, \text{ where } \lambda \in \mathbb{R}.$$

Enter the point of L with $x = 0$ and the direction $\mathbf{n}_1 \times \mathbf{n}_2$.

```
P1 = plane(Eq(2*x - y + 2*z, 6), name='Π1')
P2 = plane(Eq(4*x + 3*y - z, 2), name='Π2')

q4_3_point = ...      # the point of L where
x = 0
q4_3_direction = ...  #  $\mathbf{n}_1 \times \mathbf{n}_2$ 

verify_intersection('4.3(a) point',
q4_3_point, P1, P2, plane(Eq(x, 0)))
verify_direction('4.3(a) direction',
q4_3_direction, intersection(P1, P2))
```

§ 5. Three planes and systems of equations

Four questions, 16 marks. One point, a line of points, or none — and the letters that decide which.

5.1 — *May 2021 TZ1 Paper 2 Q6(b), 2 marks*

Consider the planes

$$\Pi_1 : 3x + 2y + z = 6$$

$$\Pi_2 : x - 2y + z = 4$$

$$\Pi_3 : 2x - y - 4z = 0$$

(b) Find the coordinates of the point where Π_1 , Π_2 and Π_3 intersect.

```
P1 = plane(Eq(3*x + 2*y + z, 6), name='Π1')
P2 = plane(Eq(x - 2*y + z, 4), name='Π2')
P3 = plane(Eq(2*x - y - 4*z, 0), name='Π3')

q5_1 = ...          # the point

verify_intersection('5.1(b)', q5_1, P1, P2,
P3)
```

5.2 — May 2022 TZ1 Paper 1 Q11(a), 4 marks

Consider the three planes

$$\Pi_1 : 2x - y + z = 4$$

$$\Pi_2 : x - 2y + 3z = 5$$

$$\Pi_3 : -9x + 3y - 2z = 32$$

(a) Show that the three planes do not intersect.

Enter what the three planes share — a point, a line or 'none' — and the value of $-9x + 3y - 2z$ along the line where Π_1 and Π_2 meet.

```
P1 = plane(Eq(2*x - y + z, 4), name='Π1')
P2 = plane(Eq(x - 2*y + 3*z, 5), name='Π2')
P3 = plane(Eq(-9*x + 3*y - 2*z, 32),
name='Π3')
```

```
q5_2 = ... # a point, a line, or
'none'
```

```
q5_2_value = ... # -9x + 3y - 2z along
the line where Π1 and Π2 meet
```

```
verify_intersection('5.2(a)', q5_2, P1, P2,
P3)
```

```
verify_find('5.2(a) value', q5_2_value,
P3.side(intersection(P1, P2).at(lam)))
```

5.3 — May 2025 TZ2 Paper 2 Q9(b), 4 marks

Consider the three planes

$$P_1 : x + y - z = -2$$

$$P_2 : 2x + by - z = 3$$

$$P_3 : x - y + 2z = d$$

where $b, d \in \mathbb{R}^+$.

The three planes intersect in a line.

(b) Find the value of b and the value of d .

```
b, d = symbols('b d')
P1 = plane(Eq(x + y - z, -2), name='P1')
P2 = plane(Eq(2*x + b*y - z, 3), name='P2')
P3 = plane(Eq(x - y + 2*z, d), name='P3')

q5_3 = [...]      # [b, d]

verify_find('5.3(b)', q5_3, [b, d],
[meet_in_line(P1, P2, P3), b > 0, d > 0])
```

5.4 — November 2025 TZ3 Paper 2 Q7, 6 marks

Consider the following two equations.

$$2x + 6y - 8z = 13 \qquad 3x - y + 3z = 12$$

(a) Determine the general solution giving the answer in parametric form.

Consider a third equation $ax + 12y - 16z = k$, where $a, k \in \mathbb{R}$.

(b) In the case where $a = 3$ and $k = -3$, determine the unique solution to the system of three equations.

(c) (i) Write down the value of a for which there is no unique solution to the system of three equations.

(ii) Hence, write down the corresponding value of k for which there is an infinite number of solutions to the system of three equations.

In (a) enter `vec(x, y, z)` in terms of `t`.

```

a, k = unknown('a'), unknown('k')
first = plane(Eq(2*x + 6*y - 8*z, 13))
second = plane(Eq(3*x - y + 3*z, 12))
third = plane(Eq(a*x + 12*y - 16*z, k))

q5_4a = ...      # vec(x, y, z) in terms of t
q5_4b = ...      # (x, y, z)
q5_4c_i = ...    # a
q5_4c_ii = ...   # k

verify_intersection('5.4(a)', q5_4a, first,
second)
verify_intersection('5.4(b)', q5_4b, first,
second, third.subs({a: 3, k: -3}))
verify_find('5.4(c)(i)', q5_4c_i, a,
[no_unique_meet(first, second, third)])
verify_find('5.4(c)(ii)', q5_4c_ii, k,
[meet_in_line(first, second, third)])

```

§ 6. The perpendicular: foot and distance

Four questions, 29 marks. From a point along the normal to the plane: where it lands, and how far that is.

6.1 — May 2022 TZ1 Paper 1 Q11(c), 6 marks

The line L , $\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}$, is the line of intersection of $\Pi_1 : 2x - y + z = 4$ and $\Pi_2 : x - 2y + 3z = 5$.

The plane Π_3 has equation $-9x + 3y - 2z = 32$.

(c) Find the distance between L and Π_3 .

Paper 1: exact.

```
L = line(vec(1, -2, 0) + lam*vec(1, 5, 3))
P3 = plane(Eq(-9*x + 3*y - 2*z, 32),
name='Π3')

q6_1 = ...      # the distance, exact

verify_distance('6.1(c)', q6_1, L, P3,
exact=True)
```

6.2 — November 2022 Paper 2

Q12(d), 9 marks

Consider the points $A(1, 2, 3)$, $B(k, -2, 1)$ and $C(5, 0, 2)$, where $k \in \mathbb{R}$.

(d) For $k \neq 9$, let Π be the plane containing A , B and C .

(i) Find the Cartesian equation of the plane Π .

(ii) Find the coordinates of the point on the plane Π which is closest to the origin $(0, 0, 0)$.

```
k = symbols('k')
Pi = plane(vec(1, 2, 3), vec(k, -2, 1),
vec(5, 0, 2))
closest = unknown('X', 3)

q6_2_i = ...      #  $\Pi$ : Eq(...)
q6_2_ii = ...     # the point closest to the
origin

verify_plane('6.2(d)(i)', q6_2_i, Pi)
verify_find('6.2(d)(ii)', q6_2_ii, closest,
[on(closest, Pi), perpendicular(closest,
Pi)])
```

6.3 — November 2023 TZ1 Paper 2

Q12(c), 7 marks

The plane P is given by $\mathbf{r} \cdot \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 7$.

Point B has position vector $\begin{pmatrix} -3 \\ 12 \\ 2 \end{pmatrix}$. A line through B perpendicular to P intersects P at point C .

(c) (i) Find the position vector of C .

(ii) Hence, find $|\overrightarrow{BC}|$.

```
plane_P = plane(Eq(dot(vec(x, y, z), vec(0,
2, -1)), 7))
B = vec(-3, 12, 2)
C = unknown('C', 3)

q6_3_i = ...      # C
q6_3_ii = ...     # |BC|

verify_find('6.3(c)(i)', q6_3_i, C, [on(C,
plane_P), perpendicular(C - B, plane_P)])
verify_distance('6.3(c)(ii)', q6_3_ii, B,
plane_P)
```

6.4 — May 2024 TZ2 Paper 1 Q11(d), 7 marks

The plane P_1 has equation $2x + 6y - 2z = 5$, and the point $A(2, \frac{1}{2}, 1)$ lies on P_1 .

The plane P_2 , parallel to P_1 , has equation $3x + 9y - 3z = -\frac{51}{2}$.

The line through A that is perpendicular to P_1 meets P_2 at the point B .

(d) (i) Find the coordinates of B .

(ii) Hence, show that the perpendicular distance between P_1 and P_2 is $\sqrt{11}$.

For (ii) enter the distance, exact.

```
A = vec(2, Rational(1, 2), 1)
P1 = plane(Eq(2*x + 6*y - 2*z, 5), name='P1')
P2 = plane(Eq(3*x + 9*y - 3*z, Rational(-51, 2)), name='P2')
B = unknown('B', 3)
```

```
q6_4_i = ...      # B
q6_4_ii = ...     # the distance between P1
and P2
```

```
verify_find('6.4(d)(i)', q6_4_i, B, [on(B,
P2), perpendicular(B - A, P1)])
verify_distance('6.4(d)(ii)', q6_4_ii, P1,
P2, exact=True)
```

§ 7. The reflection

Three questions, 16 marks. Twice the step to the foot: a reflected point, a reflected line, a parallel plane on the other side.

7.1 — November 2021 Paper 2 Q11(d), 9 marks

The line L has vector equation $\mathbf{r} = \begin{pmatrix} 0 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$, and the plane Π_3 is given by $2x - 2z = 3$.

(d) The point $B(0, -2, 0)$ lies on L .

(i) Find the reflection of the point B in the plane Π_3 .

(ii) Hence find the vector equation of the line formed when L is reflected in the plane Π_3 .

```

L = line(vec(0, -2, 0) + lam*vec(1, 1, -1))
P3 = plane(Eq(2*x - 2*z, 3), name='Π3')
B = vec(0, -2, 0)
image = unknown('image', 3)

q7_1_i = ...      # the reflection of B
q7_1_ii = ...     # the reflected line:
vec(...) + mu*vec(...)

verify_find('7.1(d)(i)', q7_1_i, image,
[reflection(image, B, P3)])
verify_line('7.1(d)(ii)', q7_1_ii, mirror(L,
P3))

```

7.2 — November 2023 TZ1 Paper 2 Q12(d), 3 marks

The plane P is given by $\mathbf{r} \cdot \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = 7$, and
point B has position vector $\begin{pmatrix} -3 \\ 12 \\ 2 \end{pmatrix}$.

(d) Find the reflection of the point B in the plane P .

```
plane_P = plane(Eq(dot(vec(x, y, z), vec(0,
2, -1)), 7))
B = vec(-3, 12, 2)
image = unknown('image', 3)

q7_2 = ...          # the reflection of B

verify_find('7.2(d)', q7_2, image,
[reflection(image, B, plane_P)])
```

7.3 — May 2024 TZ2 Paper 1 Q11(e), 4 marks

The plane P_1 has equation $2x + 6y - 2z = 5$, and the point $A(2, \frac{1}{2}, 1)$ lies on P_1 . The plane $P_2 : 3x + 9y - 3z = -\frac{51}{2}$ is parallel to P_1 at a perpendicular distance of $\sqrt{11}$ from it, and the line through A perpendicular to P_1 meets P_2 at $B(1, -\frac{5}{2}, 2)$.

(e) Find the equation of a third parallel plane P_3 which is also a perpendicular distance of $\sqrt{11}$ from P_1 .

The check builds P_3 as the mirror image of P_2 in P_1 .

```
P1 = plane(Eq(2*x + 6*y - 2*z, 5), name='P1')
B = vec(1, Rational(-5, 2), 2)

q7_3 = ...      # P3: Eq(...)

verify_plane('7.3(e)', q7_3, plane(mirror(B,
P1), P1.normal))
```



Solutions

§ 1

1.1 (a) $4 + 3 - 2 = \boxed{5}$. **(b)** $2(k^2 - 6) + 6(2k + 3) + 12 = 2(k + 3)^2 = 0$: $k = -3$; then
 $3x - 3y - 6z = q$ through A:

$$\boxed{k = -3, q = -\frac{3}{2}}. \quad \textbf{(c)} \quad (3, 9, p) = \frac{3}{2}(2, 6, -2):$$

$$\boxed{p = -3}.$$

1.2 (c) $\boxed{q = 18}$, $\boxed{r = -6}$. **(e)**
 $\boxed{\mathbf{r}_3 = (1, 8, -2) + \gamma(-4, -1, 3)}.$

§ 2

2.1 $(3, 2, 1) \times (1, -2, 1) = (4, -2, -8)$:
 $\boxed{2x - y - 4z = 0}.$

2.2 (i) $\boxed{\overrightarrow{AB} = (-3, -2, 0)}$, $\boxed{\overrightarrow{AC} = (-2, 1, -7)}$.
(ii) $(14, -21, -7)$, through A:
 $\boxed{2x - 3y - z = 6}.$

2.3 $(1, 1, 0) \times (6, 0, 6) = (6, -6, -6)$:
 $\boxed{\mathbf{n} = (1, -1, -1)}$, as a vector, not

coordinates.

2.4 $(1, 2, 1) \times (1, -1, -2) = (-3, 3, -3)$,
through R : $\boxed{x - y + z = 15}$.

2.5 $(1, 0, 1) \times (2, 1, 3) = (-1, -1, 1)$:
 $\boxed{\mathbf{n} = (1, 1, -1)}$, and $(0, 0, 2)$ gives $x + y - z = -2$.

2.6 $\overrightarrow{AB} \times \overrightarrow{AD} = (-12, 12, -12)$, through A :
 $\boxed{-x + y - z = -5}$.

2.7 (i) $\boxed{\mathbf{r} = (5, 4, 2) + \lambda(1, -1, 1) + \mu(2, 1, 3)}$.
(ii) $(5, 4, 2) \cdot (-4, -1, 3) = \boxed{-18}$, so $4x + y - 3z = 18$.

2.8 $(-1, 1, -13) \times (1, -10, 7) = (-123, -6, 9)$,
through O : $\boxed{-123x - 6y + 9z = 0}$, which is
 $41x + 2y - 3z = 0$.

§ 3

3.1 $2\lambda - 2(-\lambda) = 3$: $\boxed{\lambda = \frac{3}{4}}$, $\boxed{P\left(\frac{3}{4}, -\frac{5}{4}, -\frac{3}{4}\right)}$.

3.2 $2(2 + 3s) - (-3 + 6s) = \boxed{7}$ and $2(9 + t) - (11 + 2t) = \boxed{7}$ for every s and t .

3.3 $L : \mathbf{r} = (0, -3, 2) + \lambda(-1, 1, -1); \quad 3\lambda - 17 = 1, \lambda = 6: F(-6, 3, -4).$

§ 4

4.1 $(-7, -7, 7) = -7(1, 1, -1)$, and $(0, -2, 0)$ is on both planes.

4.2 $(2, -1, 1) \times (1, -2, 3) = (-1, -5, -3):$
 $\mathbf{r} = (1, -2, 0) + \lambda(-1, -5, -3).$

4.3 $x = 0: (0, 2, 4). (2, -1, 2) \times (4, 3, -1) = (-5, 10, 10).$

§ 5

5.1 $\left(\frac{41}{21}, -\frac{10}{21}, \frac{23}{21}\right) = (1.95, -0.476, 1.10).$

5.2 $-3x + z = -3$ and $-3x + z = 44: \text{none};$
 along $L, -9x + 3y - 2z = -15.$

5.3 $(3b - 4)y = 19 - d: b = \frac{4}{3}, d = 19.$

5.4 (a) $(x, y, z) = \left(t, \frac{27}{2} - 3t, \frac{17}{2} - 2t\right).$

(b) $(29, -73.5, -49.5).$ **(c)** $a = 4, k = 26.$

§ 6

6.1 $L \parallel \Pi_3$; from $(1, -2, 0)$ along $(-9, 3, -2)$:
 $t = \frac{1}{2}, \frac{\sqrt{94}}{2}.$

6.2 (i) $(0, k - 9, 18 - 2k) = (k - 9)(0, 1, -2)$:
 $y - 2z = -4.$ **(ii)** $t(0, 1, -2), 5t = -4$:
 $(0, -0.8, 1.6).$

6.3 (i) $22 + 5\lambda = 7, \lambda = -3$: $C(-3, 6, 5).$ **(ii)**
 $3\sqrt{5} = 6.71.$

6.4 (i) $A + \lambda(2, 6, -2)$ into P_2 : $\frac{15}{2} + 66\lambda =$
 $-\frac{51}{2}, \lambda = -\frac{1}{2}$: $B\left(1, -\frac{5}{2}, 2\right).$ **(ii)** $\overrightarrow{AB} =$
 $(-1, -3, 1), \sqrt{11}.$

§ 7

7.1 (i) $\mu = \frac{3}{8}$ to the foot, twice:

$$\left[\left(\frac{3}{2}, -2, -\frac{3}{2} \right) \right]. \quad \textbf{(ii)} \text{ through } P \left(\frac{3}{4}, -\frac{5}{4}, -\frac{3}{4} \right)$$

and B' :
$$\mathbf{r} = \left(\frac{3}{2}, -2, -\frac{3}{2} \right) + \mu(1, -1, -1).$$

7.2 $\lambda = -6$:
$$(-3, 0, 8).$$

7.3 $C = A + (A - B) = (3, \frac{7}{2}, 0)$:

$$x + 3y - z = \frac{27}{2}.$$